



EVALUATING MACHINE LEARNING MODELS FOR FOOTBALL MATCH PREDICTION AND BETTING MARKET EFFICIENCY

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Abstract

This thesis investigates whether machine learning models can outperform bookmaker odds in predicting football match outcomes and identifying profitable betting opportunities. Previous research in football forecasting has produced inconsistent conclusions, with some studies reporting profitable machine learning-based betting systems. Stübinger et al., 2019 while others were not able to outperform the bookmakers. Baboota and Kaur, 2019 This thesis therefore examines the predictive and financial performance of several machine learning approaches within a realistic football forecasting framework. The study evaluates logistic regression, decision trees, random forests, and histogram-based gradient boosting models using historical football data from eleven European football leagues stored in the European Soccer Database. The models incorporate engineered features based on team performance, rolling form statistics, ELO ratings, team ratings, and bookmaker odds. This thesis applies a chronological fixed rolling walk-forward validation framework combined with probabilistic evaluation metrics, calibration analysis, confidence analysis, and betting simulations. The results show that machine learning models are capable of learning meaningful predictive patterns from historical football data, but consistently outperforming bookmaker probabilities remains highly challenging. The histogram-based gradient boosting model achieved the strongest overall machine learning performance and generated a positive betting ROI of 1.32% under a balanced betting strategy. However, bookmaker-derived probabilities remained highly competitive across probabilistic forecasting metrics, supporting the conclusion that football betting markets remain largely information-efficient environments.

1 DATA SOURCE, ETHICS, CODE, AND TECHNOLOGY STATEMENT

Data Source: The data used in this thesis consists of historical football match data, including match outcomes, team identifiers, and bookmaker odds. The dataset originates from a publicly available source on *Kaggle*, specifically the European Soccer Database. The data is accessed through a local SQLite database file. The original ownership of the data remains with the creators and contributors of the dataset on *Kaggle*.

This thesis does not involve collecting data from human participants or animals. The dataset contains no personally identifiable information and is therefore considered anonymised. The data is used solely for academic purposes and in accordance with the terms and conditions under which it was made available on *Kaggle*.

Figures: All figures, tables, and visualisations included in this thesis were created by the author using Python. No externally copyrighted images were used, and therefore no additional permissions were required.

Code: The code used in this thesis was primarily written by the author in Python. The implementation makes use of widely used open-source libraries, including but not limited to: *pandas*, *numpy*, and *scikit-learn*.

Some parts of the code were developed with assistance from publicly available documentation and examples, as well as guidance from generative AI tools. Any reused or adapted code structures are standard implementations (such as model pipelines and evaluation procedures) and are not subject to restrictive licenses. The full code developed for this thesis is available upon request or via a repository (if applicable).

Technology: During the writing and development of this thesis, the author used several tools to support the research process. A generative language model (ChatGPT, developed by OpenAI) was used to assist with coding, debugging, and improving the clarity and structure of written text. This tool was used only to refine and support the author's original work.

For writing and typesetting, LaTeX (Overleaf) was used. Standard spell checking and grammar correction tools integrated in the writing environment were used. No additional reference management software beyond the LaTeX template was used.

Ethical Considerations: This research focuses on predictive modelling of football match outcomes and does not involve sensitive personal data. However, the application of such models in betting contexts may have ethical implications, including the potential encouragement of gambling behaviour. This thesis does not promote gambling but analyses predictive performance and market efficiency from an academic perspective. The results should therefore be interpreted cautiously and within a research context.

2 PROJECT DEFINITION, MOTIVATION & RELEVANCE

Football is the most popular sport in the world, with millions of fans following matches and a growing global industry surrounding sports betting. Predicting the outcome of football matches has long attracted attention from both practitioners and academics, given the complexity and unpredictability of the game. Currently, bookmakers provide odds that represent their estimation of match outcomes. These odds are often used as a benchmark for prediction quality, as they aggregate expert models, market sentiment, and betting activity. However, they are not perfect: odds reflect probabilities with built-in profit margins, and past research has shown that betting markets are not always fully efficient.

2.1 *societal relevance*

From a societal perspective, research into the prediction of football outcomes is highly relevant primarily because of the size and economic weight of the football betting market. Sports betting represents a multi-billion-euro industry that continues to grow each year, with football occupying the largest share worldwide. Given the scale of this market, developing a deeper scientific understanding of how match outcomes can be predicted is of clear societal interest. Improving knowledge in this domain not only contributes to academic debates, but also provides evidence-based insights into a sector that directly affects millions of consumers and involves significant financial flows.

2.2 *Scientific point of view*

From a scientific perspective, the prediction of football has been the subject of various empirical studies, but their findings have been far from conclusive. In their recent paper Stübinger et al. have demonstrated that data-driven models can yield a positive financial return when applied systematically to betting strategies, suggesting that machine learning can indeed exploit inefficiencies in bookmaker odds. (Stübinger et al., 2019) Other studies, however, found that machine learning models underperformed relative to bookmakers when evaluated using predictive accuracy metrics such as the Ranked Probability Score (RPS). (Baboota & Kaur, 2019) This divergence in outcomes indicates that the predictive potential of machine learning in football analytics is still not fully understood. The availability of a rich dataset including team and player attributes, formations, historical performance, match events, and betting odds provides a unique opportunity to re-examine this question systematically. By addressing this gap, the thesis aims to contribute to the broader scientific debate on the efficiency of prediction markets and the applicability of modern machine learning methods in sports analytics.

2.3 Research Strategy & Research Questions

The central research problem of this thesis is to investigate whether machine learning models can provide more accurate predictions of the outcomes of football matches than bookmakers' odds. To address this problem, the following research questions are proposed:

Can machine learning models that integrate player, team, formation, and historical performance features outperform bookmaker odds in predicting football match outcomes?

subquestions

- To what extent do machine learning models outperform bookmaker benchmarks on predictive performance metrics?
- Can machine learning models identify profitable betting opportunities by exploiting inefficiencies in bookmaker odds?
- How does the predictive and financial performance of machine learning models vary across different European football leagues?

3 RELATED WORK

Literature Review

In the field of machine learning applied to sports betting, Galekwa et al., 2024 is an essential reference that cannot be overlooked. This article provides a comprehensive overview of all recent scientific studies in this domain and serves as a foundational map of the research landscape. By synthesizing work across soccer, basketball, tennis, and other sports, Galekwa et al., 2024 highlight both classical approaches (e.g., logistic regression, Bayesian networks, Poisson regression) and modern methods (e.g., random forests, gradient boosting, deep neural networks). The review shows a gradual shift from bookmaker-odds-based models toward data-driven machine learning and, more recently, spatio-temporal tracking and geometric deep learning. Importantly, their synthesis also reveals inconsistencies: while some studies report predictive models that outperform bookmakers or generate positive betting returns, others find limited or no improvement over traditional benchmarks.

Several contributions illustrate these divergent outcomes. Constantinou et al., 2013 investigated inefficiencies in the European football betting market, uncovering arbitrage opportunities despite bookmakers' accuracy. In contrast, Baboota and Kaur, 2019 found that machine learning models often underperformed bookmakers when evaluated with the Ranked Probability Score (RPS). Other research demonstrated promising performance of machine learning techniques. For example, Tax and Joutstra, 2015 applied dimensionality reduction and classification algorithms to predict Dutch Eredivisie matches, achieving a

56-percent accuracy, while Stübinger et al., 2019 reported a 1.58 percent per-match return using ensemble models across five European leagues. Similarly, deep learning applications such as Atta Mills et al., 2024 showed accuracies approaching 99 percent using detailed match event data, although questions remain about overfitting and generalizability.

More recent work has expanded the scope of predictive analytics in football. Studies such as Anzer and Bauer, 2021 advanced expected goals (xG) modeling using gradient boosting, while Peters and Pacheco, 2022 examined the role of lineups in outcome prediction. Others, including Wang et al., 2024 with TacticAI and Goka et al., 2023, have introduced spatio-temporal deep learning frameworks for tactical decision-making and shot prediction. Beyond match outcomes, research has also applied machine learning to player valuation Li et al., 2022,

Together, these studies systematically cataloged by Galekwa et al., 2024 highlight both the potential and limitations of machine learning in sports betting and football analytics. On the one hand, methods such as gradient boosting, ensemble learning, and neural networks have achieved strong predictive accuracy and even profitable betting strategies. On the other hand, results vary significantly across leagues, datasets, and evaluation metrics, indicating that the field lacks consensus on which algorithms or features consistently drive predictive success. This unresolved debate underscores the relevance of the present thesis: to systematically evaluate machine learning approaches for predicting football match outcomes, compare their performance across competitions, and assess whether they can outperform benchmarks such as bookmaker odds.

4 METHOD

This section outlines the methodological framework employed throughout this thesis, including the preprocessing, feature engineering, validation, modeling, calibration, and betting evaluation stages of the research pipeline. Figure 1 presents an overview of the complete methodological process and the relationships between its individual components. Each stage of the framework was designed to ensure realistic, robust, and reproducible evaluation of football match outcome prediction and betting performance within a chronological forecasting environment.

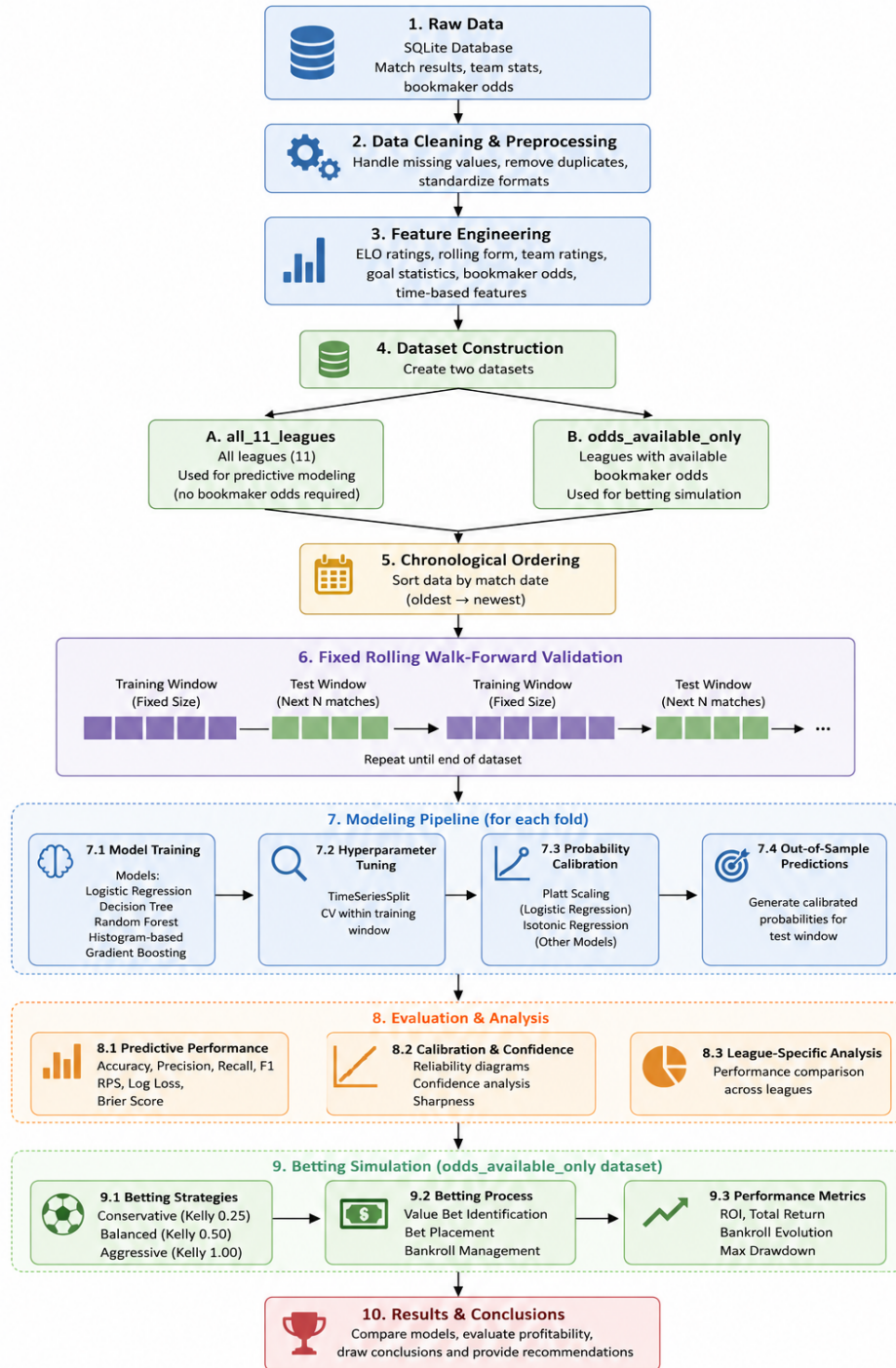


Figure 1: Flowchart of the methodological pipeline.

4.1 Dataset Description

The dataset used in this study is obtained from *Kaggle*, specifically the European Soccer Database. This dataset aggregates football data from multiple original sources and contains historical match data from several European leagues, including the English Premier League, Bundesliga, La Liga, and Serie A.

The data was originally collected from different providers. Match results, line-ups, formations, and in-game events were sourced from football-data APIs (<http://football-data.mx-api.enetscores.com/>). Bookmaker odds were obtained from <http://www.football-data.co.uk/>, which provides historical betting odds from multiple bookmakers. Additionally, team and player attributes were collected from *sofifa.com*, which is based on data from EA Sports FIFA games.

The dataset spans multiple seasons, covering matches played between approximately 2008 and 2016. In total, the dataset contains approximately 25,000 matches across 11 different leagues. Each observation represents a single football match and includes match-level and team-level information. Key variables include the date of the match, league and season identifiers, the home and away teams, and the number of goals scored by each team. Based on the goals scored, a categorical target variable is constructed representing the match outcome: home win (H), draw (D), or away win (A).

In addition, the dataset includes bookmaker odds for each possible match outcome (home win, draw, away win), collected from multiple bookmakers. These odds are used to derive implied probabilities, which serve both as input features and as a benchmark for model evaluation.

The dataset is stored in a SQLite database and accessed programmatically using Python.

Overall, the dataset combines match results, betting market data, and team attributes, providing a comprehensive basis for modelling football match outcomes.

4.2 Exploratory Data Analysis

4.2.1 Subsets

The distribution of match outcomes differs across the eleven league-specific subsets, highlighting the heterogeneous nature of the dataset. Although home wins are the most common outcome in all leagues, the relative frequency of home wins, draws, and away wins varies between competitions, reflecting differences in playing style, competitive balance, and tactical approaches.

Importantly, each league represents an independent competition in which teams only compete against other teams within the same league. This implies that there is no direct interaction between subsets, as teams from different leagues do not play against each other in the dataset. As a result, each league can be viewed as a separate data-generating process, with its own underlying dynamics and statistical properties.

4.2.2 *Distribution of match outcome*

The distribution of match outcomes is moderately imbalanced. Home wins represent the largest share of observations, accounting for 45.87% of all matches. Away wins occur in 28.74% of matches, while draws account for the remaining 25.39%.

This distribution reflects the well-established home advantage in football, where teams playing at home are more likely to win due to factors such as crowd support, familiarity with the playing environment, and reduced travel fatigue. The relatively lower proportion of away wins further highlights the difficulty of achieving positive results in away matches.

From a modeling perspective, this imbalance has important implications. A naïve model that always predicts a home win would already achieve an accuracy close to 46%, which underscores the limitation of accuracy as a sole evaluation metric. Therefore, this study emphasizes the use of probabilistic evaluation measures such as the Ranked Probability Score (RPS), log loss, which provide a more nuanced assessment of predictive performance

4.3 *Data Preprocessing*

4.3.1 *Missing values*

An analysis of bookmaker odds revealed that missing values are not uniformly distributed across the dataset, but instead occur at two distinct levels: structural and incidental.

At the structural level, a complete absence of bookmaker odds is observed for specific leagues. In particular, the Poland Ekstraklasa and the Switzerland Super League contain no bookmaker odds for any matches. This indicates that the missingness is not random, but driven by a lack of market coverage for these competitions. As a result, these leagues cannot be included in analyses that rely on bookmaker information, such as betting simulations or direct comparisons with bookmaker predictions. Despite this limitation, these leagues are retained in the full dataset used for model training and general performance evaluation. Excluding them entirely would reduce the diversity of

the dataset and limit the ability of the model to learn from a broader range of competitive environments. Retaining these subsets ensures that the model is exposed to a wider variety of match dynamics, thereby improving generalizability. In addition to structural missingness, a small number of matches within leagues with bookmaker coverage were found to lack odds for all bookmakers. These cases represent incidental missing data rather than systematic absence.

Since bookmaker odds are required to compute expected value and simulate betting strategies, matches without odds cannot be used in the betting evaluation. Therefore, these observations are excluded from the betting and benchmarking analysis.

However, these matches are retained in the full dataset used for model training and general performance evaluation. This approach ensures that no potentially informative observations are discarded unnecessarily, while maintaining a realistic and valid betting simulation.

4.3.2 *Chronological ordering*

Since football match prediction is inherently a time-dependent problem, maintaining the temporal order of observations is essential throughout the analysis. All matches in the dataset are therefore sorted chronologically based on their match date before any preprocessing, feature engineering, or model training steps are performed.

This chronological ordering is particularly important because the study aims to simulate a realistic forecasting environment in which predictions are generated using only information that would have been available prior to the match. Without preserving the temporal structure of the data, information from future matches could unintentionally influence feature construction or model training, leading to data leakage and overly optimistic performance estimates.

4.4 *Feature engineering*

4.4.1 *Rolling features*

To capture short-term team performance and recent form, a set of rolling performance features is constructed for both the home and away team. These features are calculated over the previous five matches played by each team and are intended to represent the recent competitive state of a team prior to the target match. The rolling features used in this study are:

- Home points and away points

- Home goal difference and away goal difference
- Home average goals scored and away average goals scored
- Home average goals conceded and away average goals conceded
- Home win rate and away win rate
- Home clean sheets and away clean sheets
- Home rest days and away rest days

In addition to these absolute measures, difference-based features are constructed to directly compare both teams:

- Difference in points
- Difference in goal difference
- Difference in average goals scored
- Difference in average goals conceded
- Difference in win rate
- Difference in rest days

A rolling window of five matches is selected to balance responsiveness and stability. Using a relatively short window allows the features to adapt to recent changes. At the same time, the window is sufficiently large to reduce the influence of random variation from individual matches.

All rolling statistics are calculated using lagged observations. This means that the current match is excluded from the rolling calculations, ensuring that only information available prior to the match is used for prediction.

4.4.2 *Elo Features*

In addition to rolling performance statistics, Elo-based features are constructed to capture long-term team strength. The Elo rating system dynamically updates team ratings after each match based on expected and observed outcomes, allowing team strength to evolve over time.

- Home Elo rating
- Away Elo rating
- Elo difference
- Home Elo win probability

The home and away Elo ratings represent the estimated strength of both teams prior to the match. The Elo difference captures the relative strength between teams, while the home Elo win probability represents the implied probability of a home win based on the Elo model.

A home advantage parameter is incorporated into the Elo calculation to account for the well-documented advantage associated with playing at home. Elo ratings are updated sequentially after each match, ensuring that only historical information is used.

4.4.3 Bookmaker Features

For matches with available bookmaker odds, bookmaker-derived features are constructed to capture market-based expectations of match outcomes. Since multiple bookmakers are available in the dataset, the average odds across bookmakers are calculated separately for the home win, draw, and away win outcomes.

The average odds are used to construct bookmaker-implied probabilities. This is done by taking the inverse of the average odds and normalizing the resulting values across the three possible outcomes. This normalization accounts for the bookmaker margin and ensures that the implied probabilities sum to one.

- Implied home win probability
- Implied draw probability
- Implied away win probability

Average bookmaker odds are used for feature construction because they provide a more stable representation of market expectations than the odds of a single bookmaker. They reduce the influence of individual outliers and better reflect the aggregated information contained in the betting market. In contrast, the best available odds across bookmakers are retained for the betting simulation. This distinction is important because a bettor would rationally select the bookmaker offering the highest payout when placing a bet. Therefore, average odds are used to represent market expectations in the predictive model, while best odds are used to simulate realistic betting execution.

4.5 Experimental Procedure

4.5.1 Dual dataset strategy

Due to the heterogeneous availability of bookmaker odds, the experimental setup uses two separate datasets. This distinction is necessary to balance two objectives: maximizing the amount of data available for predictive modeling while enabling fair comparison with bookmaker expectations and realistic betting simulations.

The first dataset contains all matches across the eleven leagues and is used for the independent evaluation of the machine learning models. This dataset does not include bookmaker-derived features and allows the models to learn from the full range of available observations. By retaining all leagues, including those without bookmaker coverage, the analysis preserves dataset diversity and enables the evaluation of generalization across different competitive environments.

The second dataset consists only of matches for which bookmaker odds are available. This subset is used for performance-

oriented models that incorporate bookmaker probability features, as well as for betting simulations and benchmarking against bookmaker predictions.

The separation between these datasets is methodologically important. A central objective of this study is to evaluate whether machine learning models can outperform bookmaker expectations and potentially identify profitable betting opportunities. In this context, bookmaker odds serve not only as an additional source of information, but also as a benchmark that the models aim to outperform.

For this reason, bookmaker-derived features are excluded from the independent evaluation models. Including bookmaker probabilities in these models would create a circular comparison, as the models would directly rely on the same market information they are intended to compete against. Such an approach would make it difficult to determine whether predictive improvements originate from the machine learning models themselves or simply from embedded bookmaker knowledge.

Consequently, the independent models are trained exclusively on team, performance, and contextual features. This setup allows the analysis to evaluate whether machine learning techniques can independently extract predictive patterns and potentially obtain an informational edge over bookmakers.

At the same time, bookmaker-derived probabilities are included in the performance-oriented models that aim to maximize predictive accuracy. In this context, bookmaker information is treated as an additional predictive signal rather than a benchmark competitor.

In addition, betting-related analyses require bookmaker odds to compute implied probabilities, expected value, and potential returns. Matches without bookmaker odds cannot be meaningfully included in such analyses, since no betting decision can be simulated for those observations.

By maintaining both datasets, the study ensures that all available data is used efficiently while preserving the validity and fairness of bookmaker comparisons and betting evaluations.

4.5.2 *Dual Walk-Forward Validation*

Given the temporal nature of football match prediction, a walk-forward validation strategy is used to evaluate model performance. Traditional random train-test splits are not appropriate for this type of data, as they may introduce data leakage by allowing information from future matches to influence the training process. Such leakage would result in overly optimistic performance estimates and unrealistic evaluation conditions.

To address this issue, the models are evaluated using a sequential walk-forward framework in which predictions are always generated on future matches that were not available during training. The dataset is first ordered chronologically based on match date, after which an initial training window is used to fit the models. Predictions are then generated for subsequent matches in temporal order.

Rather than continuously expanding the training data, a fixed-size rolling training window is applied throughout the evaluation procedure. After every 279 observations, corresponding to approximately three weeks of football matches across the included leagues, the models are retrained using only the most recent historical matches contained within the fixed training window. Older observations falling outside the window are discarded from the retraining process. This rolling-window approach allows the models to adapt more effectively to temporal changes in football competitions, such as evolving team strength, tactical developments, transfers, and league dynamics, while simultaneously limiting the influence of potentially outdated historical information.

Hyperparameter optimization is performed separately within each chronological training window. Consequently, all hyperparameter tuning remains fully isolated from the subsequent out-of-sample test observations, ensuring that the evaluation framework remains free from future information leakage.

A dual walk-forward setup is implemented to align with the dual dataset strategy described previously. The first walk-forward pipeline evaluates the independent models on the full dataset without bookmaker-derived features. This setup assesses whether machine learning models can independently identify predictive patterns and potentially outperform bookmaker expectations.

The second walk-forward pipeline is applied to the subset of matches with available bookmaker odds and includes bookmaker-derived probability features. This performance-oriented setup is used to maximize predictive accuracy and to evaluate betting strategies based on expected value.

4.5.3 *Model selection*

The selection of machine learning models in this study is guided by prior research in football prediction and sports analytics, as discussed in the literature review. Existing studies demonstrate that predictive performance in football forecasting varies substantially across algorithms, datasets, and evaluation procedures, with no clear consensus regarding the consistently best-performing approach. Consequently, this study adopts

a comparative modeling framework that includes both interpretable baseline models and more advanced ensemble-based techniques.

4.5.3.1 LOGISTIC REGRESSION

Logistic regression is included as a baseline model due to its simplicity, interpretability, and strong performance in classification tasks. Previous studies have shown that logistic regression remains competitive in football outcome prediction, particularly when combined with carefully engineered features such as team strength indicators and recent form.

In addition, logistic regression produces probabilistic outputs that are well suited for evaluation using metrics such as log loss and the Ranked Probability Score (RPS). Regularization is applied to reduce the risk of overfitting and improve generalization performance.

4.5.3.2 DECISION TREE

Decision trees are included as a simple non-linear model capable of capturing interaction effects between features. Unlike linear models, decision trees can model hierarchical decision boundaries and complex feature relationships without requiring explicit feature transformations.

4.5.3.3 RANDOM FOREST

Random forests extend decision trees by aggregating predictions across multiple trees trained on randomized subsets of the data. This ensemble approach reduces variance and generally improves robustness and generalization performance.

Random forests are particularly suitable for football prediction because they can capture non-linear relationships and feature interactions while remaining relatively resistant to overfitting. Prior literature has shown that ensemble methods frequently outperform single-model approaches in sports analytics tasks.

4.5.3.4 HISTOGRAM-BASED GRADIENT BOOSTING

Histogram-based gradient boosting models are included because boosting techniques have consistently demonstrated strong performance on structured tabular datasets. Gradient boosting iteratively improves predictions by focusing on errors made by previous trees, enabling the model to capture subtle performance patterns and complex interactions between features.

Compared to traditional gradient boosting implementations, histogram-based boosting is computationally more efficient and scales better to larger datasets. The literature review further highlights that boosting approaches frequently achieve state-of-the-art results in football analytics and expected goals modeling.

4.6 *Hyperparameter Tuning*

The hyperparameters of the evaluated machine learning models were determined separately within each chronological training window. For every retraining period, hyperparameter optimization was performed exclusively using the historical observations available within the corresponding training data, while the subsequent test period remained completely unseen. This approach ensured that the tuning process reflected realistic forecasting conditions and prevented information leakage from future matches into model selection.

4.7 *Evaluation Metrics*

To evaluate predictive performance, multiple complementary evaluation metrics are used. Since football match prediction is a multi-class classification problem with imbalanced outcome distributions, relying solely on classification accuracy would provide an incomplete assessment of model quality. Therefore, both classification-based and probabilistic evaluation metrics are considered.

4.7.1 *Accuracy*

Classification accuracy is included as a general measure of predictive correctness, representing the proportion of matches for which the predicted outcome matches the observed result. However, because home wins occur substantially more frequently than draws or away wins, accuracy alone may overestimate model performance.

4.7.2 *Ranked probability score*

To address this limitation, probabilistic evaluation metrics are emphasized. The Ranked Probability Score (RPS) is used as the primary evaluation metric because it evaluates the full predicted probability distribution across the three possible match outcomes. Unlike simple accuracy measures, RPS penalizes both incorrect predictions and poorly calibrated probability estimates.

4.7.3 *Log Loss*

In addition, log loss is used to assess the quality and calibration of predicted probabilities. Log loss assigns larger penalties to highly confident but incorrect predictions, making it particularly useful for evaluating probabilistic forecasting models.

4.7.4 *Brier score*

The Brier score is also included as a measure of probabilistic accuracy. This metric evaluates the squared difference between predicted probabilities and actual outcomes, providing an interpretable measure of calibration quality.

4.8 *Betting Evaluation Framework*

In addition to traditional predictive evaluation metrics, model performance is also assessed within a betting framework. This allows the analysis to evaluate not only whether the models generate accurate predictions, but also whether these predictions can identify potentially profitable betting opportunities relative to bookmaker odds.

For matches with available bookmaker odds, the predicted probabilities generated by the machine learning models are compared to the implied probabilities derived from bookmaker odds. The bookmaker-implied probabilities are calculated by taking the inverse of the decimal odds and normalizing the resulting values across all possible outcomes to account for the bookmaker margin.

Betting decisions are based on the concept of expected value (EV). The expected value of a bet is calculated as:

$$EV = (p \times o) - 1 \quad (1)$$

where p represents the predicted probability of a particular outcome and o denotes the corresponding decimal bookmaker odds.

A positive expected value indicates that the predicted probability exceeds the probability implied by the bookmaker odds, suggesting a potentially profitable betting opportunity.

Rather than extensively optimizing betting parameters directly on historical performance, the betting evaluation framework uses a set of predefined betting strategy profiles with different levels of risk exposure. Specifically, conservative, balanced, and aggressive betting strategies are evaluated. These strategies differ in terms of expected value thresholds, Kelly staking fractions, and betting restrictions such as minimum confidence levels and accepted odds ranges.

This approach was chosen to reduce the risk of overfitting the betting framework toward historical test results while simultaneously improving the realism of the evaluation setting. By defining the betting strategies prior to model evaluation and applying them consistently across all evaluated models, the resulting betting performance comparisons remain more robust and directly comparable.

To determine stake sizes, a fractional Kelly Criterion strategy is applied. The Kelly framework allocates larger stakes to bets with higher estimated advantage while limiting excessive exposure to risk. A fractional version of the Kelly strategy is used to reduce volatility and improve robustness under the high uncertainty conditions typically associated with football betting markets.

The betting simulation uses the best available odds across bookmakers for each possible outcome. This reflects realistic betting behavior, as rational bettors would generally select the bookmaker offering the highest available payout.

Betting performance is evaluated using return on investment (ROI), total profit, number of bets placed, hit rate, and final bankroll value. Importantly, betting simulations are restricted to matches with available bookmaker odds, ensuring that all betting decisions reflect realistic market conditions.

4.9 *Methodological Design Choices and Alternatives*

Several methodological alternatives were considered throughout the design of this thesis. The final methodological framework was selected to balance predictive realism, robustness, interpretability, and practical applicability within football forecasting and betting environments.

One of the most important methodological decisions concerned the validation strategy. Traditional random train-test splits and shuffled cross-validation approaches were not used, as these methods risk introducing information leakage by allowing future match information to indirectly influence the training process. Instead, a chronological fixed rolling walk-forward validation framework was implemented. This approach ensures that all predictions are generated strictly out-of-sample using only information that would have been available at the time of prediction, thereby better reflecting realistic forecasting conditions.

A fixed rolling training window was selected instead of an expanding window approach. While expanding windows benefit from larger training datasets and potentially more stable parameter estimation, they also incorporate increasingly older observations that may become less relevant over time due to tactical evolution, changing team dynamics, transfers, and broader struc-

tural changes within football competitions. The fixed rolling framework was therefore chosen to maintain a stronger focus on more recent football patterns while still preserving sufficient training data for model stability.

Although recent studies have increasingly explored deep learning methods, spatio-temporal neural networks, and advanced tracking-based architectures, these approaches were not included within the present study. Deep learning models typically require substantially larger datasets, highly granular event data, or player tracking information in order to generalize effectively. Given the structure and scale of the available dataset, the use of more traditional machine learning approaches was considered more appropriate and methodologically robust.

Similarly, Bayesian networks and Poisson-based models were not selected as primary approaches within this research framework. While these methods have historically played an important role in football analytics and match outcome prediction, the focus of this thesis was specifically directed toward comparing modern machine learning techniques capable of handling larger numbers of engineered features and more complex non-linear interactions within a unified evaluation framework.

The selected methodological framework was therefore designed to balance interpretability, robustness, computational feasibility, and predictive complexity. This comparative setup enabled the study to evaluate whether more advanced ensemble-based machine learning approaches provide meaningful improvements over simpler statistical models, while simultaneously assessing whether such models can realistically compete with bookmaker-based forecasting benchmarks under probabilistic and betting-oriented evaluation settings.

5 RESULTS

5.1 *Introduction tot the results*

In this chapter, the empirical results of the machine learning models are presented and evaluated. The analysis focuses on both predictive performance and betting performance across multiple European football leagues. Model performance is assessed using several classification and probabilistic forecasting metrics, including accuracy, F1-score, Ranked Probability Score (RPS), log loss, and Brier score. In addition, the financial performance of the models is evaluated through a simulated expected-value betting strategy based on bookmaker odds.

5.2 Baseline

The bookmaker baseline was included as the primary benchmark throughout the evaluation process. , bookmaker implied probabilities provide a realistic reference point. Bookmaker odds aggregate large amounts of market information and expert knowledge and are therefore a strong benchmarks.

The bookmaker baseline achieved relatively moderate predictive performance across both classification-based and probabilistic forecasting metrics, including accuracy, Ranked Probability Score (RPS), log loss, and Brier score. Although bookmaker probabilities incorporate large amounts of market information and expert knowledge, the obtained results suggest that accurately predicting football match outcomes remains inherently difficult.

In particular, the classification accuracy of slightly above 52% demonstrates that even highly information-efficient betting markets are unable to consistently achieve very high levels of categorical prediction performance. This reflects the substantial uncertainty present in football matches,

Similarly, the probabilistic forecasting metrics indicate that substantial uncertainty persists even when estimating outcome probabilities rather than only predicting the final class label. The RPS, log loss, and Brier score results collectively suggest that assigning highly accurate and well-calibrated probabilities to football outcomes remains a challenging task, even for market-based probability estimates.

Table 1: Overall bookmaker baseline performance on the odds-available dataset

Model	Accuracy	F1 Macro	RPS	Log Loss	Brier
Bookmaker Baseline	0.5296	0.3862	0.3934	0.9731	0.5792

5.3 Predictive performance analysis

5.3.1 overall classification performance

Table 2 presents the overall out-of-sample predictive performance of the evaluated machine learning models across the different evaluation settings

Across all evaluation settings, the performance differences between the evaluated machine learning models remained relatively modest. Overall, the relatively small performance differences between the evaluated approaches suggest that football outcome prediction remains a highly constrained forecasting problem.

A particularly notable finding is the strong overall performance of the histogram-based gradient boosting model across the evaluated predictive and probabilistic forecasting metrics. Across nearly all evaluation settings, the model achieved the strongest RPS, log loss, and Brier score values among the evaluated machine learning approaches. This suggests that the boosting approach was more effective at estimating outcome probabilities and capturing the uncertainty inherently present within football match outcomes.

At the same time, logistic regression consistently achieved the highest F1 macro scores across all evaluation settings. Since F1 macro equally weights all outcome classes, this indicates that logistic regression produced more balanced predictive performance across home wins, draws, and away wins. In contrast, the stronger overall probabilistic forecasting performance of histogram-based gradient boosting appears to originate primarily from improved probability estimation rather than substantially improved class balance.

The histogram-based gradient boosting model also achieved the highest classification accuracy within both the full eleven-league dataset and the odds-available subset. These results indicate that the model was able to simultaneously improve both categorical prediction performance and probabilistic forecasting quality.

The results therefore suggest that different evaluation metrics reward different predictive characteristics. While histogram-based gradient boosting achieved stronger overall probability estimation performance, logistic regression appeared to generate more balanced predictions across the three outcome classes. This distinction is particularly relevant within betting-oriented forecasting tasks, where reliable probability estimation and class-level predictive behavior may both influence decision-making performance.

Including bookmaker odds as input features substantially improved predictive performance across all evaluated models. This suggests that bookmaker odds contain highly informative predictive signals that are difficult to fully replicate using historical football data alone.

Despite these improvements, the bookmaker baseline itself remained highly competitive across all probabilistic forecasting metrics. This indicates that bookmaker markets continue to represent a difficult benchmark to outperform, even when bookmaker odds are incorporated directly into the machine learning models.

Table 2: Overall out-of-sample predictive performance across evaluation settings after dynamic hyperparameter tuning

Evaluation Set	Model	Accuracy	F1 Macro	RPS	Log Loss	Brier
All 11 Leagues	HistGradientBoosting	0.5146	0.3785	0.4055	0.9960	0.5942
	Random Forest	0.5121	0.3623	0.4071	0.9972	0.5953
	Logistic Regression	0.5034	0.4065	0.4103	1.0071	0.6015
	Decision Tree	0.5063	0.3870	0.4130	1.0435	0.6043
Odds Available Only	HistGradientBoosting	0.5226	0.3858	0.4014	0.9883	0.5889
	Random Forest	0.5183	0.3681	0.4030	0.9898	0.5902
	Logistic Regression	0.5111	0.4134	0.4054	0.9981	0.5955
	Decision Tree	0.5142	0.3918	0.4082	1.0373	0.5982
With Odds Features	HistGradientBoosting	0.5246	0.3937	0.3956	0.9785	0.5825
	Random Forest	0.5246	0.3787	0.3967	0.9792	0.5832
	Logistic Regression	0.5180	0.4266	0.3989	0.9874	0.5880
	Decision Tree	0.5180	0.3956	0.4024	1.0164	0.5914

5.3.2 Performance across leagues

Table 4 and the complete league-level results presented in Appendix A show that substantial performance differences emerged across the evaluated football leagues, suggesting that the predictability of football match outcomes varies considerably between competitions. In general, the differences observed between leagues were larger than the differences observed between the evaluated machine learning models themselves. This indicates that league-specific characteristics may have a stronger influence on forecasting difficulty than the specific machine learning algorithm being applied.

The strongest overall predictive performance was observed in the Portugal Liga ZON Sagres, the Spanish LIGA BBVA, and the Netherlands Eredivisie. Across these competitions, both the bookmaker baseline and the evaluated machine learning models achieved relatively high classification accuracies together with substantially lower RPS, log loss, and Brier score values compared to the other leagues. For example, the bookmaker baseline achieved an accuracy of 0.5594 and an RPS of 0.3784 in Spain, while Portugal produced the overall lowest probabilistic forecasting errors across multiple evaluation metrics. These findings suggest that match outcomes within these leagues may follow more stable and predictable competitive patterns.

In contrast, substantially weaker predictive performance was observed in competitions such as the Poland Ekstraklasa and the Switzerland Super League. These leagues consistently generated lower classification accuracies and considerably worse probabilistic forecasting metrics across nearly all evaluated models. In Poland, RPS values exceeded 0.43 for all evaluated approaches, while log loss values frequently surpassed 1.05. Similarly, Switzerland produced some of the weakest probabilistic forecasting results within the dataset. The elevated RPS,

log loss, and Brier score values indicate substantially greater uncertainty in the estimated outcome probabilities within these leagues.

The results further demonstrate that the relative performance patterns of the evaluated machine learning models remained relatively stable across competitions. Histogram-based gradient boosting generally achieved the strongest overall probabilistic forecasting performance among the evaluated machine learning approaches, particularly in leagues such as Italy, Belgium, Portugal, and the Netherlands. Logistic regression, however, consistently achieved the strongest F1 macro scores across many leagues, including Belgium, England, Portugal, and Scotland. This suggests that logistic regression produced more balanced predictive performance across home wins, draws, and away wins, even when its overall probability estimation performance was weaker.

Finally, the bookmaker baseline remained highly competitive across nearly all leagues where bookmaker odds were available. In most cases, the bookmaker probabilities achieved the strongest overall probabilistic forecasting performance, producing the lowest RPS, log loss, and Brier score values within the respective competitions. This further reinforces the earlier finding that bookmaker markets remain difficult forecasting benchmarks to outperform consistently.

5.4 *Classification and error analysis*

5.4.1 *confusion matrix analysis*

Figure 2 presents the normalized confusion matrices for the evaluated forecasting models. Several clear and consistent classification patterns emerge across all approaches. Most notably, all evaluated models experienced substantial difficulty correctly identifying draw outcomes. In nearly all cases, the majority of actual draws were instead classified as home wins, indicating a strong systematic bias toward predicting home victories.

This tendency is particularly visible for the bookmaker baseline, random forest, and histogram-based gradient boosting models, which almost never predicted draws correctly. For example, the bookmaker baseline correctly classified virtually none of the observed draw outcomes, while the random forest model similarly showed near-zero draw recall. Instead, these models overwhelmingly classified draw matches as home wins. This behavior suggests that maximizing overall predictive accuracy may encourage the models to prioritize the most common and comparatively easier-to-predict outcome classes.

In contrast, logistic regression displayed a noticeably different classification behavior. Although its overall probabilistic forecasting performance was weaker than that of histogram-based gradient boosting, logistic regression predicted draws more frequently and achieved substantially higher draw recall compared to the other evaluated models. This finding helps explain the consistently stronger F1 macro scores previously observed for logistic regression, as F1 macro equally weights performance across all outcome classes. However, while these differences in draw prediction behavior were clearly noticeable, the absolute differences between the evaluated models remained relatively modest overall. The confusion matrices therefore suggest that all evaluated approaches struggled with fundamentally similar classification challenges.

Across all evaluated approaches, home wins represented the easiest outcome category to predict correctly. Most models achieved home win recall values above 0.80, indicating that home victories were identified relatively reliably. Away wins were more difficult to classify, but still substantially easier than draws. A large proportion of away wins were nevertheless incorrectly classified as home wins, further reinforcing the dominant home-win prediction bias observed throughout the evaluated forecasting systems.

Overall, the confusion matrices reveal that the evaluated models primarily struggle with distinguishing draws from the other outcome classes. The results therefore suggest that football outcome prediction is not only difficult because of overall uncertainty, but also because certain outcome categories—particularly draws—are inherently much harder to separate from the surrounding match patterns using the available historical features.

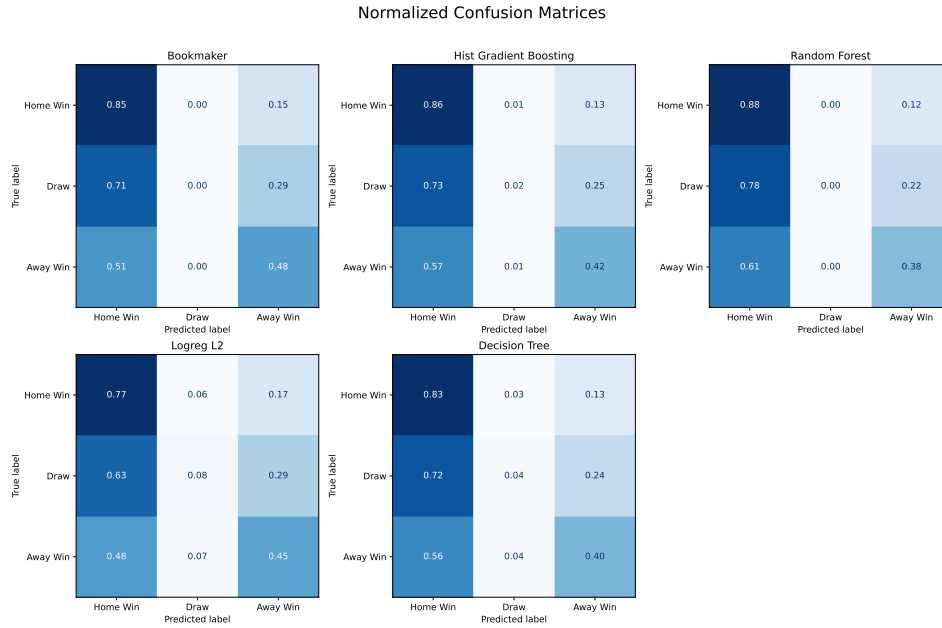


Figure 2: Normalized confusion matrices for the evaluated forecasting models.

5.4.2 prediction confidence and calibration analysis

Figure 3 presents the relationship between prediction confidence and observed forecasting accuracy across the evaluated models. Overall, a clear positive relationship emerged between prediction confidence and actual predictive performance. Higher confidence predictions generally corresponded with higher observed accuracy, indicating that the evaluated models were capable of capturing meaningful uncertainty information within their probability estimates.

Despite the overall positive relationship between prediction confidence and observed accuracy, noticeable differences in confidence reliability behavior emerged between the evaluated models. especially on higher confidence levels. The histogram-based gradient boosting model remained closely aligned with the perfect confidence reliability reference line across almost the entire confidence range, indicating particularly strong consistency between predicted confidence and actual forecasting performance.

The bookmaker baseline and the random forest model frequently performed slightly above the perfect confidence reliability line at higher confidence intervals. This indicates that the observed prediction accuracy occasionally exceeded the corresponding confidence estimates, suggesting a mildly conservative or underconfident prediction behavior at elevated probability levels.

In contrast, logistic regression and the decision tree model more frequently fell below the perfect confidence reliability line at higher confidence levels. For these models, predicted confidence occasionally exceeded the observed forecasting accuracy, indicating a tendency toward mild overconfidence for highly certain predictions.

Although these differences in confidence reliability behavior were clearly noticeable, the absolute deviations from the perfect reliability reference line remained relatively modest overall. Consequently, all evaluated approaches still demonstrated a generally meaningful relationship between prediction confidence and realized forecasting accuracy.

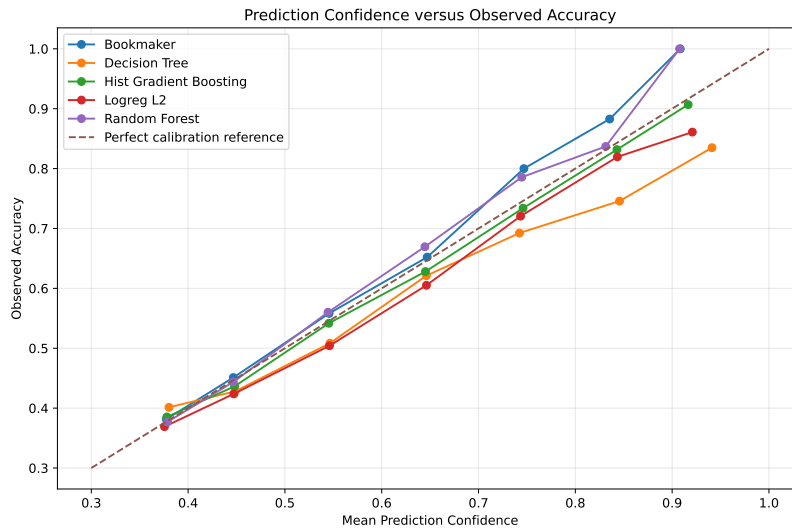


Figure 3: Relationship between prediction confidence and observed accuracy across the evaluated forecasting models.

Additional confidence distribution plots presented in figure 4 further illustrate the relationship between prediction confidence and forecasting correctness. Across all evaluated models, correct predictions generally received higher confidence scores than incorrect predictions. However, substantial overlap between the confidence distributions of correct and incorrect predictions remained visible. Incorrect predictions were therefore not limited to only low-confidence forecasts, but also occurred at moderate and occasionally relatively high confidence levels. This finding further emphasizes the substantial uncertainty that remains inherently present within football outcome prediction tasks

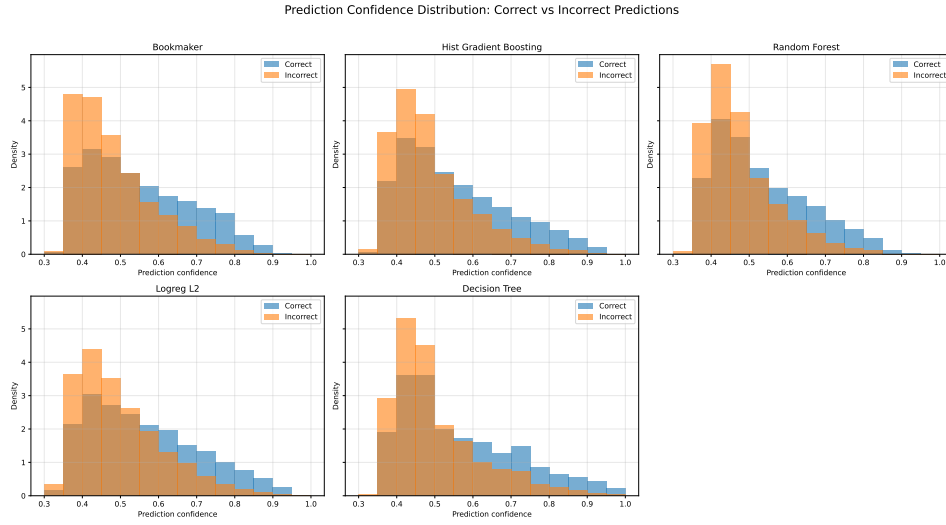


Figure 4: Prediction confidence distributions for correct and incorrect predictions across the evaluated forecasting models.

The outcome-specific confidence reliability curves presented in figure 5 reveal substantial differences between the three outcome categories. Home win predictions and away win consistently demonstrated the strong confidence reliability behavior, with all evaluated models remaining relatively close to the ideal confidence reference line across a broad range of probability levels. In contrast, draw predictions displayed substantially weaker reliability behavior. The evaluated models don't assign high confidence to draw outcomes, and the relationship between predicted confidence and observed outcome frequency remained considerably less stable.

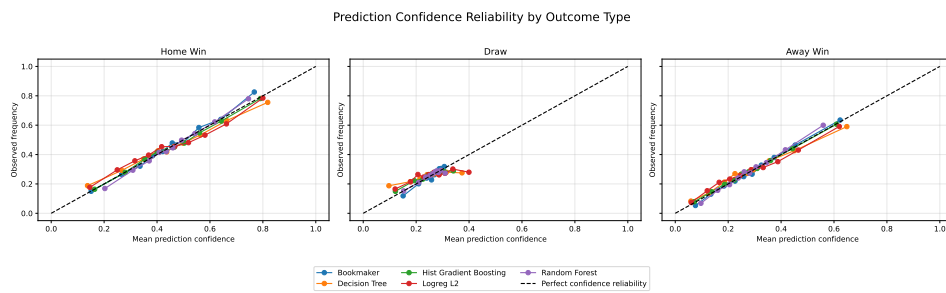


Figure 5: Prediction confidence reliability curves by outcome type for the evaluated forecasting models.

5.5 *Betting performance analysis*

Table 3 presents the betting performance of the evaluated forecasting models across the predefined conservative, balanced, and aggressive betting strategies. Overall, the results demonstrate that strong predictive performance does not automatically translate into profitable betting performance. While several models achieved relatively competitive classification and probabilistic forecasting metrics, consistently generating positive betting returns remained substantially more difficult.

The strongest overall betting performance was achieved by the histogram-based gradient boosting model under the balanced betting strategy. This combination generated a positive ROI of 1.32% and produced the highest overall profit and final bankroll value among all evaluated machine learning approaches. Importantly, this performance was achieved across a relatively large number of bets, suggesting that the results were not solely driven by a small number of highly volatile betting outcomes.

In contrast, the aggressive betting strategies consistently produced extremely poor results across nearly all evaluated machine learning models. Although these strategies generated substantially larger numbers of bets, the corresponding bankrolls rapidly declined toward zero. This suggests that the aggressive betting configurations exposed the models too heavily to the inherent uncertainty and volatility present within football betting markets. The results therefore indicate that even when predictive models identify positive expected value opportunities, excessive betting frequency and overly aggressive stake sizing can quickly eliminate profitability.

The conservative betting strategies generally reduced overall losses compared to the aggressive approaches, but still failed to consistently generate positive long-term returns. This highlights the difficulty of identifying sufficiently large and stable market inefficiencies within football betting markets, even when relatively restrictive betting conditions are applied.

A particularly important finding is that betting profitability did not always align with traditional predictive accuracy measures. Several model-strategy combinations achieved relatively high hit rates while still producing negative ROI values. This demonstrates that correctly predicting match outcomes alone is insufficient for profitable betting performance. Instead, profitability depends on the interaction between prediction accuracy, probability calibration, bookmaker odds, and stake sizing decisions.

The bookmaker baseline itself generated very few betting opportunities under the predefined betting constraints, particularly within the balanced strategy configuration. This behavior is

logically consistent, as bookmaker-implied probabilities are directly derived from the bookmaker odds themselves, leaving limited opportunities for identifying positive expected value bets against the market baseline.

Overall, the betting results further reinforce the earlier probabilistic forecasting findings. Models with stronger probability estimation quality and more reliable calibration behavior generally produced more stable betting performance than models that focused primarily on maximizing categorical prediction accuracy alone.

A particularly notable observation is that many model-strategy combinations still achieved hit rates close to or above 50%, especially under the balanced and conservative betting strategies, while nevertheless remaining unprofitable overall. This finding illustrates an important characteristic of sports betting markets: achieving a relatively high proportion of correct predictions alone is insufficient for generating positive betting returns.

Table 3: Betting performance across predefined betting strategies and forecasting models

Model	Strategy	Bets	Hit Rate	Profit	ROI	End Bankroll	Total Staked
Bookmaker	Aggressive	36	0.5278	263.68	0.0568	10263.68	4643.07
Bookmaker	Balanced	1	0.0000	-132.66	-1.0000	9867.34	132.66
Decision Tree	Balanced	3611	0.5289	-332.67	-0.0009	9667.33	359810.15
Decision Tree	Conservative	1456	0.5412	-841.67	-0.0077	9158.33	109096.57
Decision Tree	Aggressive	7732	0.4736	-9994.51	-0.0636	5.49	157255.98
HistGradientBoosting	Balanced	2660	0.5376	6839.43	0.0132	16839.43	516289.50
HistGradientBoosting	Conservative	763	0.5347	-3219.95	-0.0526	6780.05	61211.82
HistGradientBoosting	Aggressive	6072	0.4728	-9999.23	-0.0751	0.77	133232.19
Logistic Regression	Conservative	1304	0.5337	-2329.68	-0.0191	7670.32	122194.65
Logistic Regression	Balanced	3667	0.5097	-8232.16	-0.0345	1767.84	238408.75
Logistic Regression	Aggressive	6965	0.4696	-9999.94	-0.0599	0.06	166852.34
Random Forest	Balanced	1563	0.5195	-4535.16	-0.0257	5464.84	176779.93
Random Forest	Conservative	377	0.5438	-1163.37	-0.0329	8836.63	35384.45
Random Forest	Aggressive	4690	0.4377	-9999.97	-0.1206	0.03	82938.61

6 DISCUSSION

6.1 Answering the Research Questions

The results of this thesis provide several important insights into the application of machine learning for football match prediction and betting-related forecasting. In response to the research questions posed in this study, the following conclusions can be drawn.

RQ1.1: To what extent do machine learning models outperform bookmaker benchmarks on predictive performance metrics?

The results indicate that bookmaker benchmarks remained highly competitive across nearly all evaluated predictive perfor-

mance metrics. While the evaluated machine learning models achieved forecasting performance levels relatively close to the bookmaker benchmark under several evaluation settings, none of the models consistently outperformed the bookmaker baseline across the overall experiments.

The histogram-based gradient boosting model consistently achieved the strongest overall machine learning performance across both classification accuracy and probabilistic forecasting metrics, particularly when bookmaker-derived probability features were included. Among the evaluated machine learning approaches, this model achieved the highest overall accuracy and the strongest probabilistic forecasting performance in terms of RPS, log loss, and Brier score. In contrast, logistic regression achieved comparatively strong F1-macro performance, suggesting more balanced predictive behavior across the different outcome classes.

The relatively modest performance differences observed between increasingly complex machine learning approaches further suggest that a substantial proportion of the predictive structure within football match outcomes may already be captured through relatively simple statistical relationships. At the same time, the consistently strong bookmaker benchmark demonstrates the difficulty of outperforming markets that already aggregate large volumes of public information, expert knowledge, and collective market behavior.

RQ1.2: Can machine learning models identify profitable betting opportunities by exploiting inefficiencies in bookmaker odds?

The betting simulations provide limited but noteworthy evidence that machine learning models may occasionally identify profitable betting opportunities under carefully controlled betting conditions. The strongest betting performance was achieved by the histogram-based gradient boosting model under the balanced betting strategy, which generated a positive return on investment across a relatively large number of bets.

However, the overall betting results demonstrate that generating stable long-term betting profits remains extremely difficult. Many model-strategy combinations achieved hit rates close to or above 50% while nevertheless remaining unprofitable overall. This finding demonstrates that predictive accuracy alone is insufficient for profitable sports betting. Instead, profitability depends on the interaction between probability calibration, bookmaker pricing efficiency, expected value estimation, and bankroll management.

The aggressive betting strategies consistently produced severe bankroll declines across nearly all evaluated models. This suggests that even when machine learning models identify positive expected value opportunities, the uncertainty and volatility inherent within football betting markets remain sufficiently large

to eliminate profitability under overly aggressive betting behavior.

Overall, the findings suggest that while limited market inefficiencies may exist, these inefficiencies appear relatively small and difficult to exploit consistently. The bookmaker market therefore continues to represent an extremely efficient and difficult benchmark to outperform in practical betting environments.

RQ1.3: How does the predictive and financial performance of machine learning models vary across different European football leagues?

Substantial performance variation was observed across the evaluated European football leagues. In general, the differences between leagues were larger than the differences between the evaluated machine learning models themselves. This suggests that league-specific characteristics may exert a stronger influence on forecasting difficulty than the specific modeling approach that is applied.

The strongest predictive performance was generally observed in leagues such as the Portuguese Liga ZON Sagres, the Spanish LIGA BBVA, and the Netherlands Eredivisie. In contrast, substantially weaker predictive performance emerged in competitions such as the Poland Ekstraklasa and the Switzerland Super League. These leagues consistently produced lower classification accuracy and weaker probabilistic forecasting metrics across all evaluated models.

The findings indicate that football forecasting difficulty is not uniform across competitions and may be strongly influenced by league structure, tactical variability, competitive balance, scoring patterns, and overall data consistency. This further suggests that league-specific modeling approaches may potentially provide stronger forecasting performance than generalized multi-league models in future research.

Main Research Question: Can machine learning models that integrate player, team, formation, and historical performance features outperform bookmaker odds in predicting football match outcomes?

Overall, the findings of this thesis suggest that machine learning models are capable of learning meaningful predictive patterns from historical football data and can achieve forecasting performance levels broadly comparable to bookmaker benchmarks. However, consistently outperforming bookmaker odds remains highly challenging.

The inclusion of bookmaker-derived probability features substantially improved predictive performance across all evaluated machine learning models, indicating that bookmaker odds themselves contain highly informative predictive signals that are difficult to replicate using historical football data alone. At the same

time, the bookmaker baseline itself remained extremely competitive across both predictive and betting-related evaluation metrics.

Consequently, the results support the conclusion that football betting markets remain largely information-efficient environments. While machine learning models can contribute meaningful probabilistic forecasting insights and may occasionally identify limited market inefficiencies, consistently and robustly outperforming bookmaker odds appears substantially more difficult than maximizing traditional predictive performance metrics alone.

6.2 *Contribution to Existing Literature*

The findings of this thesis contribute to the ongoing debate within the football forecasting literature regarding whether machine learning models can consistently outperform bookmaker benchmarks and generate profitable betting strategies. Similar to Baboota and Kaur (2019), the results demonstrate that bookmaker-derived probabilities remained difficult to outperform. These findings reinforce the conclusion that bookmaker markets remain highly information-efficient and that machine learning models struggle outperform the bookmakers.

the betting simulations provide partial support for studies reporting limited profitable betting opportunities using machine learning approaches. Stübinger et al. (2019), for example, reported a positive betting return of approximately 1.58% using ensemble-based football betting strategies. Similarly, the histogram-based gradient boosting model evaluated in this thesis achieved a positive ROI of 1.32% under the balanced betting strategy. Although the profitability observed in this thesis remained relatively modest, the similarity in ROI magnitude suggests that small positive betting returns may be achievable under carefully controlled betting conditions.

However, the findings of this thesis additionally demonstrate that such profitability remains highly unstable and difficult to maintain consistently. Many evaluated betting strategies produced substantial losses despite relatively strong predictive hit rates, suggesting that betting profitability depends not only on prediction accuracy, but also on probability calibration, market efficiency, and bankroll management.

6.3 *Research Limitations and Methodological Constraints*

Several limitations and methodological constraints should be considered when interpreting the findings of this thesis.

First, the study relies exclusively on historical football match data and structured pre-match information. Although the dataset includes team performance indicators, ELO ratings, bookmaker odds, and engineered form-related features, many potentially influential factors were not available within the dataset. Examples include player injuries, suspensions, weather conditions, managerial changes, transfer activity, dressing-room dynamics, and real-time tactical adjustments. Since football match outcomes are influenced by a large number of dynamic and partially unpredictable factors, the absence of such contextual information may limit the overall predictive performance of the evaluated machine learning models.

Second, the betting simulations were conducted under several simplifying assumptions that may not fully reflect real-world betting environments. The simulations assume continuous access to the best available bookmaker odds without restrictions, transaction costs, liquidity limitations, stake limitations, or account restrictions imposed by bookmakers. In practice, profitable bettors may face reduced betting limits or restricted market access, which could substantially reduce the practical profitability of the evaluated betting strategies.

Another important limitation concerns the probabilistic stability of betting profitability. While certain model-strategy combinations generated positive betting returns, profitability remained highly sensitive to betting strategy configuration, bankroll management, and model calibration quality. The relatively small positive ROI values observed in some experiments may therefore partially reflect variance or favorable betting sequences rather than stable long-term market inefficiencies. This limitation is particularly relevant given the inherently noisy and uncertain nature of football match outcomes.

The league structure of the dataset also introduces several methodological constraints. Not all evaluated leagues contained complete bookmaker odds coverage, resulting in separate evaluation settings for general forecasting and betting-related analysis. While this dual evaluation framework allowed the inclusion of a broader set of football matches for predictive modeling, it may reduce direct comparability between certain experimental settings.

computational limitations influenced several methodological design choices within the study. The repeated retraining procedure required by the chronological walk-forward validation framework substantially increased computational complexity and training time. As a result, the number of evaluated models, hyperparameter configurations, and retraining frequencies remained necessarily constrained. Although the selected framework was designed to balance realism, robustness, and computa-

tional feasibility, alternative validation procedures or larger-scale hyperparameter optimization may potentially produce different results.

6.4 *Broader Societal and Scientific Implications*

This thesis contributes to the growing field of machine learning applied to football forecasting and sports betting by providing a probabilistic evaluation framework that combines predictive modeling, calibration analysis, betting simulation, and bookmaker comparison across multiple European football leagues.

The results demonstrate that machine learning models are capable of learning meaningful predictive patterns from historical football data, but also highlight the substantial difficulty of consistently outperforming bookmaker markets. In particular, the findings emphasize the importance of probabilistic forecasting quality, calibration behavior, and uncertainty estimation rather than relying exclusively on traditional classification accuracy measures.

From a societal perspective, the findings are relevant given the rapid growth of online sports betting markets and the increasing accessibility of artificial intelligence tools. Although certain betting strategies generated small positive returns under controlled conditions, the results also demonstrate the substantial volatility and financial risk associated with aggressive betting behavior. This highlights the importance of realistic expectations regarding the practical capabilities of machine learning systems within gambling environments.

Finally, the findings suggest that machine learning-based football forecasting systems may still provide practical value as probabilistic decision-support tools for analysts, researchers, and betting-related applications, particularly when combined with robust risk management and calibration-focused evaluation frameworks.

6.5 *Recommendations for Future Research*

First, future studies could incorporate more advanced data sources such as player tracking data, event-stream data, injury reports, transfer information, and tactical positioning metrics. The inclusion of richer contextual information may improve the ability of machine learning models to capture the dynamic and complex nature of football match outcomes.

Second, future research could further explore advanced machine learning architectures such as deep neural networks, graph neural networks, transformer-based models, and ensemble learning

frameworks. Although the relatively traditional machine learning approaches evaluated in this thesis achieved competitive performance, more advanced architectures may potentially extract additional predictive patterns from large-scale football datasets.

Finally, Future studies may also benefit from focusing on league-specific forecasting systems rather than generalized multi-league models. The substantial variation in predictive performance observed across the evaluated European football leagues suggests that competition-specific characteristics may require more specialized modeling approaches.

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APPENDIX A

Table 4: League-level out-of-sample predictive performance across evaluated models

League	Model	Accuracy	F1 Macro	RPS	Log Loss	Brier
Belgium Jupiler League	Bookmaker	0.5214	0.3768	0.3995	0.9845	0.5870
	HistGradientBoosting	0.5157	0.3737	0.4071	0.9978	0.5960
	Random Forest	0.5132	0.3555	0.4094	1.0018	0.5988
	Logistic Regression	0.5231	0.4233	0.4121	1.0044	0.6015
	Decision Tree	0.5017	0.3658	0.4140	1.0454	0.6065
England Premier League	Bookmaker	0.5281	0.3867	0.3959	0.9826	0.5841
	HistGradientBoosting	0.5088	0.3767	0.4074	1.0033	0.5976
	Random Forest	0.5140	0.3707	0.4049	0.9970	0.5942
	Logistic Regression	0.5145	0.4278	0.4072	1.0077	0.5995
	Decision Tree	0.5004	0.3881	0.4146	1.1125	0.6085
France Ligue 1	Bookmaker	0.5052	0.3689	0.4013	1.0060	0.6035
	HistGradientBoosting	0.4937	0.3657	0.4088	1.0207	0.6131
	Random Forest	0.4863	0.3424	0.4120	1.0251	0.6161
	Logistic Regression	0.4758	0.3969	0.4124	1.0314	0.6205
	Decision Tree	0.4941	0.3792	0.4147	1.0634	0.6206
Germany 1. Bundesliga	Bookmaker	0.5139	0.3665	0.4116	0.9904	0.5902
	HistGradientBoosting	0.5147	0.3706	0.4176	1.0020	0.5975
	Random Forest	0.5071	0.3526	0.4155	0.9975	0.5947
	Logistic Regression	0.4935	0.3850	0.4243	1.0169	0.6069
	Decision Tree	0.5011	0.3788	0.4261	1.0334	0.6070
Italy Serie A	Bookmaker	0.5296	0.3964	0.3916	0.9783	0.5839
	HistGradientBoosting	0.5254	0.3847	0.3993	0.9947	0.5945
	Random Forest	0.5201	0.3750	0.4022	0.9972	0.5962
	Logistic Regression	0.5033	0.3995	0.4047	1.0084	0.6040
	Decision Tree	0.5192	0.3978	0.4039	1.0207	0.6011
Netherlands Eredivisie	Bookmaker	0.5466	0.3956	0.3889	0.9597	0.5708
	HistGradientBoosting	0.5436	0.3997	0.3951	0.9735	0.5784
	Random Forest	0.5349	0.3790	0.3968	0.9735	0.5791
	Logistic Regression	0.5218	0.3885	0.4003	0.9814	0.5853
	Decision Tree	0.5310	0.3922	0.4023	0.9881	0.5868
Poland Ekstraklasa	HistGradientBoosting	0.4426	0.3052	0.4374	1.0596	0.6384
	Random Forest	0.4619	0.3031	0.4360	1.0558	0.6364
	Logistic Regression	0.4220	0.3320	0.4519	1.0881	0.6564
	Decision Tree	0.4234	0.3121	0.4544	1.1162	0.6614
Portugal Liga ZON Sagres	Bookmaker	0.5466	0.4051	0.3691	0.9359	0.5540
	HistGradientBoosting	0.5464	0.4214	0.3743	0.9472	0.5599
	Random Forest	0.5426	0.3933	0.3787	0.9521	0.5641
	Logistic Regression	0.5420	0.4663	0.3774	0.9534	0.5656
	Decision Tree	0.5331	0.4111	0.3835	1.0274	0.5722
Scotland Premier League	Bookmaker	0.5095	0.3762	0.4086	0.9870	0.5879
	HistGradientBoosting	0.5058	0.3961	0.4155	0.9957	0.5947
	Random Forest	0.4971	0.3608	0.4202	1.0052	0.5999
	Logistic Regression	0.4971	0.4351	0.4166	0.9970	0.5957
	Decision Tree	0.4971	0.4052	0.4242	1.0201	0.6086
Spain LIGA BBVA	Bookmaker	0.5594	0.3951	0.3784	0.9335	0.5513
	HistGradientBoosting	0.5502	0.3898	0.3888	0.9535	0.5642
	Random Forest	0.5489	0.3790	0.3906	0.9559	0.5658
	Logistic Regression	0.5397	0.3996	0.3951	0.9698	0.5732
	Decision Tree	0.5450	0.4011	0.3939	1.0070	0.5710
Switzerland Super League	HistGradientBoosting	0.4841	0.3499	0.4287	1.0329	0.6198
	Random Forest	0.4795	0.3380	0.4328	1.0369	0.6228
	Logistic Regression	0.4860	0.3801	0.4319	1.0427	0.6252
	Decision Tree	0.4916	0.3992	0.4334	1.0448	0.6254